Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, ME, V
Name of the Paper: Design of Machine Element-I
Paper Code: TME-502

Time: 1.5 hour Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. Explain the various phase of design, and what are the factors influencing the design in mechanical engineering. CO 1

OR

b. Write the note on different materials used in engineering application also write about their mechanical properties. CO 1

Q2. (10 Marks)

a. Write in detail about stress concentration factor and its cause, how you will account for stress concentration in design of machine parts. CO 2

OR

b. Prove that by distortion energy theory "yield stress in shear is 0.577 of yield stress in tension". CO 3

Q3. (10 Marks)

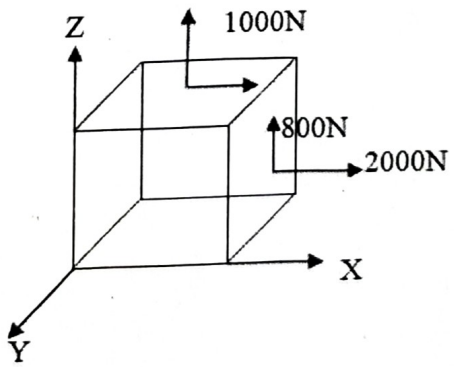
a. Discuss Tresca-Guest theory of failure and construct the region of safety. CO 3

OR

b. A solid circular shaft of diameter 45mm is loaded by bending moment 650N-m, torque 900N-m and an axial tensile force of 30KN. the shaft material is ductile with yield strength of 280MPa. Determine the factor of safety according to Tresca and Von-mises theory of failure. CO 3

Q4. (10 Marks)

a. A cube of 5 mm side is loaded as shown in figure, determine the principle stress, will the cube yield if the yield strength of the material is 70 Mpa, use Von mises theory. CO 3



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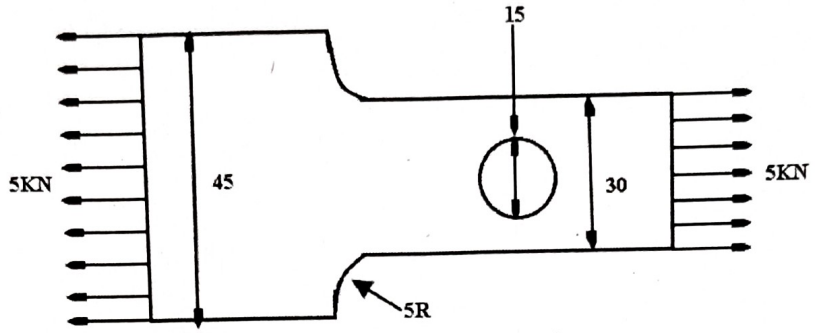
OR

b. A flat plate subjected to a tensile force of 5KN is shown in figure 2, the plate material is grey cast iron FG200 and the factor of safety is 2.5 determine the thickness of plate. Use data book where required. CO 2



(10 Marks)

Q5.

a. Discuss about the endurance limit, endurance life, endurance strength, draw the SN curve and explain low cycle fatigue and high cycle fatigue. CO 3

OR

b. A steel rod is subjected to a reversed axial load of 180KN. find the diameter of the rod for factor of safety of 2. Neglect column action. the material has an ultimate tensile strength of 1070MPa and yield strength of 910Mpa. the endurance limit in reversed bending may be assumed to be one -half of the ultimate tensile strength. other correction factor is as: for axial loading =0.7, for machined surface=0.8 for size=0.85, for stress concentration=1.0. CO 3

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